Joseph Li

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SKILLS

Languages & Databases: Python, TypeScript, Go, SQL, PostgreSQL, Redis, C

Infrastructure & Development: Linux, Docker, Kubernetes, Git, GitHub Actions, CI/CD, AWS Bedrock, React, Next.js

Systems & Data: API Gateways, Data Pipelines, Learning-to-Optimize, Regression Testing, Observability, Asynchronous Programming, Benchmarking, PyTorch, NumPy, Pandas

PROFESSIONAL EXPERIENCE

CORTEX AI **Menlo Park, CA**

AI Engineer Intern ***Jun. 2026 - Aug. 2026***

- Designed and launched an LLM API gateway in **Go**, **PostgreSQL**, and **Redis** that routes OpenAI, Anthropic, and **AWS Bedrock** through one endpoint for a 2,000-seat customer and meters $10K in monthly model spend
- Built a pre-release test suite with **GitHub Actions**, expanding coverage from 5 smoke tests to 159 tasks so tool regressions were caught before release
- Established live monitoring in **Langfuse**, adopted by all 9 engineers, tracking per-session latency and failures and routing each failure to the engineer responsible
- Reduced wasted model spend by stopping runaway agent loops and using an **LLM judge** to flag unnecessary tool calls

YOYO MARKETS INC. **New York, NY (Remote)**

AI Engineer ***Aug. 2025 - Dec. 2025***

- Automated manual research and editorial work with a **Python** and LLM generation pipeline, producing **50+ markets per week at 5x the hand-curated rate**
- Developed an **asynchronous Python** scraper tracking 150+ high-signal Twitter accounts in real time, feeding that pipeline **3x the signal volume** of public trend aggregators
- Devised a ranking step that uses pairwise LLM comparisons and Elo ratings to screen 200+ candidate markets per cycle and send the strongest to founder review
- Shipped a **Next.js** review dashboard on **Vercel** that founders use daily, cutting **publishing turnaround from 30 minutes to under 5 minutes**

RESEARCH AND PROJECTS

EFFICIENT LLMS | *PyTorch*, *CUDA*, *LoRA* **Jan. 2026 - May 2026 | USC**

- Converted **Llama 3.2-1B** into a **Mixture-of-Experts** model with **LoRA** adapters and kept held-out quality level with the original, perplexity **9.60 versus 9.66**
- Quantized weights from **FP16** to **INT4**, cutting model size **61%** while keeping generated answers coherent

POWER GRID DISPATCH MODEL | *Python* **Jan. 2026 - May 2026 | USC**

- Engineered a **learning-to-optimize** pipeline for New York's grid, where a model learns generation costs from public load, price, and outage data and an optimizer sets the hourly plan
- Cut the extra cost above the cheapest possible hourly plan **by 47%** versus the old schedule by adding transmission limits and a reserve margin

POSTGRESQL BUFFER MANAGER | *C*, *PostgreSQL* **Jan. 2025 - May 2025 | USC**

- Replaced **PostgreSQL**'s default memory rule with a first-in, first-out rule that discards the oldest unused data
- Benchmarked its performance against PostgreSQL's default implementation and found the simpler rule **matched its speed** at 1MB of memory

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA **Los Angeles, CA**

M.S. Computer Science | GPA: 3.6/4.0 *Sep. 2024 - May 2026*

Relevant Coursework: ML Accelerators, Operating Systems

UNIVERSITY OF MICHIGAN, ANN ARBOR **Ann Arbor, MI**

B.S.E. Data Science | GPA: 3.7/4.0 *Sep. 2020 - May 2024*